

## Important properties

- ① Ductility.
- ② Ability of material to be drawn into wires.  
%age of elongation or %age of reduction in area.  
% elongation is 15%.
- ③ Plasticity  $\rightarrow$  ability of a material to undergo permanent deformation.  
$$\text{deformation} = \text{ED} + \text{PD}$$
- ④ Resiliance  $\rightarrow$  ability to absorb energy up to the elastic limit. strain energy absorbed / unit vol  
 $\rightarrow$  spring  
Modulus of resiliance
- ⑤ Toughness  $\rightarrow$  it is the area under stress strain curve, upto fracture.  
 $\rightarrow$  impact load.
- ⑥ Malleability  $\rightarrow$  Beaten into thin sheets.  
 $\rightarrow$  compressive force.  
 $\rightarrow$  All malleable materials are not ductile.
- ⑦ Brittleness  $\rightarrow$  very little or no plastic deformation.  
% elongation  $\leq$  5%

Hardness  $\rightarrow$  to resist deformation, penetration, indentation etc.

$\rightarrow$  can be initiated by surface treatments

Stiffness  $\rightarrow$  ability of the material to resist deformation under the action of external load within the elastic limit.

$\rightarrow$  modulus of elasticity is the measure of stiffness.

(b) Creep  $\rightarrow$  if load is applied for long period of time. it undergoes permanent deformation. at a high temp deformation phenomenon. eg  $\rightarrow$  gas turbine blades.



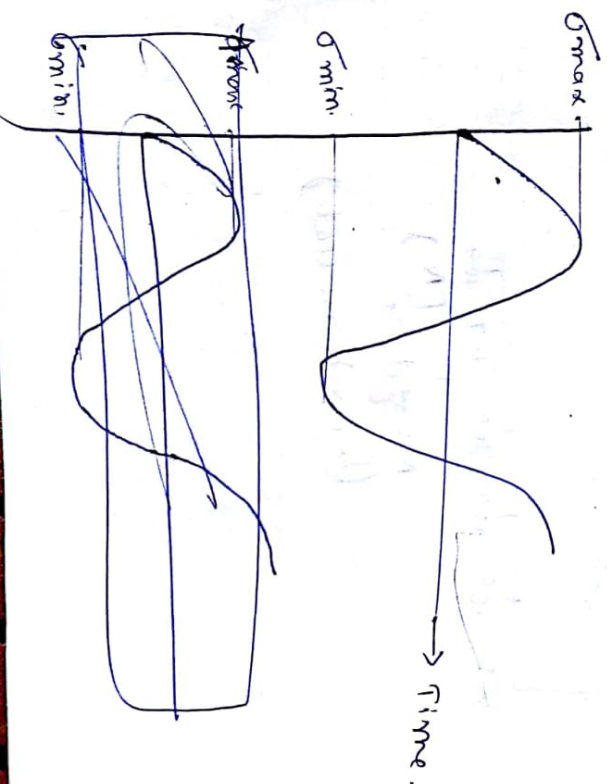
It is the balance between work hardening and recovery processes.

Instantaneous deformation is mainly elastic.

Primary (transient creep)  $\rightarrow$  slope of strain vs time ~~decreases~~

Creep mechanism  $\rightarrow$  diffusion, dislocation, grain boundary sliding

Fatigue cyclic stresses.





## Factors that affect fatigue life.

- magnitude of stress
- quality of the surface.
- Introducing compressive stress
- optimising geometry

## Creep can be minimized by

- high melting temp
- high elastic modulus
- large grain size.
- stainless steel, refractory metals, superalloys

## SELECTION OF MATERIALS

- 1) cost.
- 2) availability.
- 3) Mech properties
- 4) Manufacturability

## Imp metals

- ① Ferrous
- ② Nonferrous -

Pg 453  $\rightarrow$  yield stress.  
 $Fe, FeE$

Pg 462  $\rightarrow$  tensile strength yield stress elongation of break  
 $Fe200$

Based on chemical composition.

Pg 463  
Properties of plain carbon steel.

C35  
Pg 455 Alloy steel  $\rightarrow$  low & medium alloy steel, high alloy steel  
alloy tool steel.

Pg 466  
cast iron  $\rightarrow$  Pg 458

grey cast iron  $\rightarrow$  459

Material of component  
 $\rightarrow$  homogenous & isotropic  
Reliability  
high for cost  
high for high service conditions  
high for tough working conditions.

### Factors of safety.

The degree of uncertainty of strength, reliability & service condition etc.

The uncertainty regarding the exact cost of failure (change in material properties during service. Stress levels to which localised stress can be developed.

$$n = \frac{\text{Failure stress}}{\text{Design stress}} = \frac{\text{Failure load}}{\text{Design load}}$$

$\rightarrow$  high factor safety for impact loads  
 $\rightarrow$  low for static loads, magnitude & direction of static loading force unpredictable.

Brittle material (eg. steel)

$$n = \frac{\sigma_y}{\sigma_d}$$

Brittle material  $\rightarrow$   $\frac{\sigma_y}{K_t K_{ts}}$   
 $\downarrow$   
static concentration factors.

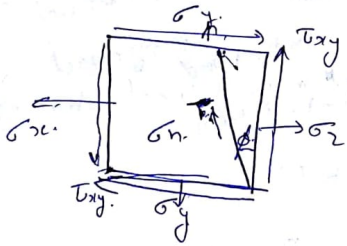


## Fatigue loading

importance of material H.L.

$$\eta = \frac{\sigma_{en} \times \text{correction factors}}{\sigma_{y \text{ add.}}}$$

fatigue stress concentration factor



## Theories of failure

(5)

when a member is subjected to biaxial or triaxial state of stress, prediction of failure becomes difficult due to the following reasons.

- all the normal stress components influence yielding.
- a large no of complex multiaxial test are required.
- influence of outside factors such as stress concentration, temp and environment. all these condition can cause the testing procedure costly, time consuming and sometime difficult to achieve.

In order to overcome these difficulties, theories have been developed, that relates material behaviour in complex situation to the behaviour in a simple uniaxial tension or compression test. By some characterise modulus such as stress, strain, energy.

1st theory - max principal stress theory. (pg 22) eq-2.8(a)

If  $\sigma_1 > \sigma_y$ , failure of specimen subjected to any combination of loads occurs when the max principal stress is equal to yield or ultimate strength.

assuming  $\sigma_1 > \sigma_2$

★  
② Max shear stress theory (pg 22, eq-2.8c)

$$\frac{\sigma}{\sigma_0} = \sigma_1 - \sigma_2$$

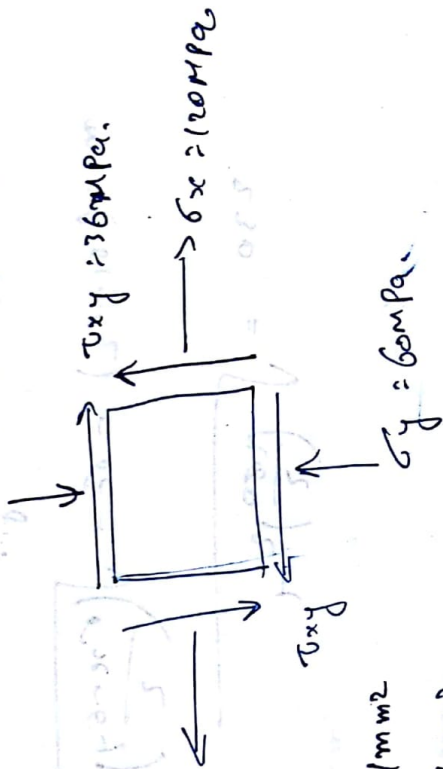
③ max distortional energy or shear energy theory (eq-2.8.0)

The strain energy of material in unit volume is equal to.

$$\frac{\sigma_e}{n} = \sqrt{\frac{\sigma_1^2 + \sigma_2^2}{2}} - \sigma_1 \sigma_2$$

① stoichiometry  
② total energy theory.

[illegible]



$$\sigma_x = 120 \text{ N/mm}^2$$

$$\sigma_y = -60 \text{ N/mm}^2$$

$$\tau_{xy} = 36 \text{ N/mm}^2$$

$$\sigma_c = 230 \text{ N/mm}^2$$

$$\sigma_c = \left( \frac{\sigma_x + \sigma_y}{2} \right) + \sqrt{\left( \frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^2}$$

$$\frac{230}{n} = \left( \frac{120 - 60}{2} \right) + \sqrt{\left( \frac{120 + 60}{2} \right)^2 + (36)^2}$$

$$\frac{230}{n} = \frac{60}{2} + \sqrt{(90)^2 + (36)^2}$$

$$n = 1.811$$

$$FOS = 1.811$$



max shear stress theory:

$$\max(\sigma_1 - \sigma_2) = \sigma_c = \sqrt{\left(\sigma_x - \sigma_y\right)^2 + 4\tau_{xy}^2}$$

$$230 = \sqrt{\left(\frac{180}{2}\right)^2 + \tau_{xy}^2}$$

Q → A bolt is subjected to tensile load of 18 kN and a shear load of 12 kN. Yield stress = 330 MPa.  $\sigma_{os} = 2.5$ . Find the dia of the bolt. by any two sol<sup>n</sup> of failure

$$\sigma_x = 18 \text{ kN}$$

$$\tau_{xy} = 12 \text{ kN}$$

$$\sigma_c = 330 \text{ MPa} = 330 \text{ N/mm}^2$$

$$n = 2.5$$

from max stress theory

$$\sigma_c = \frac{1.8 \times 10^3}{A}$$

$$\sigma_y = 0$$

$$\tau = \tau_{xy} = \frac{1.2 \times 10^3}{A}$$



so,  $\sigma_x = \frac{\sigma_z}{2.5}$

$\frac{132}{181103} = \frac{1}{A}$

$A = 136.36$   
 $A = 90.9 \text{ mm}^2$

from shear stress theory

$\tau_{xy} = 132 = \frac{1210}{A}$

$A = 90.9 \text{ mm}^2$

(log 2)

$\frac{1}{2} = \frac{1}{2}$

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a) A rod is subjected to bending moment of 32 Nm and a twisting moment of 2 kNm. Find the dia of the rod. yield stress = 40 MPa.  $n = 3$ .

sol

$$T(BM) = 3 \times 10^3 \text{ Nm}$$

$$T.M = 2 \times 10^3 \text{ Nm}$$

$$\sigma_c = \sigma_{\text{tensile}}$$

$$n = 3$$

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$$\frac{M}{I} = \frac{\sigma}{c}$$

$$\frac{M}{\frac{\pi d^4}{32}} = \frac{\sigma}{\frac{d}{2}}$$

$$\sigma = \frac{32M}{\pi d^3}$$

$$\sigma = \frac{40}{3} = \frac{32 \times 3 \times 10^3}{\pi d^3}$$

$$d^3 = 60.2 \text{ mm}$$



eg- 1.3 (b)

$$\frac{T}{S} = \frac{\tau}{r}$$

$$\frac{2 \times 10^6}{2000}$$

a) A cylindrical boiler of dia 2m. thickness = 20mm,  
internal pressure 1.5 MPa, yield strength 350 MPa.  
find factor of safety.

sol/ Pg-99 7.1 (b)

$$\sigma_t = \frac{P d_i}{2 t}$$

$$\sigma_t = \frac{1.5 \times d_i}{2 \times 20}$$

$$\sigma_t = \frac{1.5 \times 2000}{2 \times 20}$$

$$75$$

$$\sigma_t = 75 \text{ N/mm}^2$$

fos = .4-66

7.1 (c).

pg(20)

Stress concentrations are generated throughout a

is applied to any situation. results in values of actual stress being higher than that calculated by means of the usual equations from elementary stress of material. In many machine elements. The effect of cross section may not be uniform and other geometric discontinuities may be present. Such as holes, splines, keyways, fillets etc. There are some of the location of stress concentration. At these locations the stress distribution is non uniform and in the neighbourhood of each discontinuity the stress is much higher than the rest of section.

$$K_t = \frac{\sigma_{max}}{\sigma_{nom}}$$

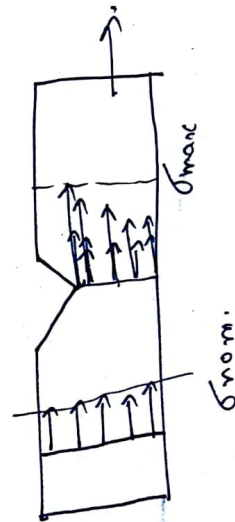
$$K_t \geq \frac{\sigma_{max}}{\sigma_{nom}}$$

### Causes of stress concentration.

1) variation in properties of material from point to point in a member e.g. cracks, cavities, air holes, nonmetallic inclusions

2) pressure at point at which loads or members are applied. e.g. cam & follower, gear teeth, contact h/w wheel & rail.

→ Abrupt changes of cross section. This includes oil holes, grooves etc.



Theoretical stress concentration factor depends only on the geometry.